

□ 22 Paired samples test

Example 1

A group of 10 subjects are randomly selected from a population to participate in a weight loss program. For each subject their before and after weights in Kg are as tabulated below. Is the weight loss program effective?

	1	2	3	4	5	6	7	8	9	10
Before	77.9	76.5	78.3	81.5	79.3	80.3	78.6	77.7	75.5	77.9
after	74.4	72.3	76.4	83.8	78.1	79.3	72.1	78.0	72.9	76.0

At first glance, a t test might appear to be another problem where the t test applies – we have a scale variable (*weight*) and a binary variable (we might call it *time_point* with two levels – before and after)

While using a t test (or what might be more fully called an independent samples t test) is not incorrect, it may not be the optimal way to proceed. There is clearly a relationship, or pairing, between the various *before* and *after* values – they pertain to the same person.

E.g. the third subject weighted 78.3 kg before the program and the same person weighed 76.4 kg after,

Note that we did not have this pairing in any of the previous examples we looked at using the (independent samples) t test.

For example, recall our problem (partially reproduced opposite) examining if first year male and female students paid different amounts of rent.

Male	92.18	94.52	:
Female	108.52	117.12	:

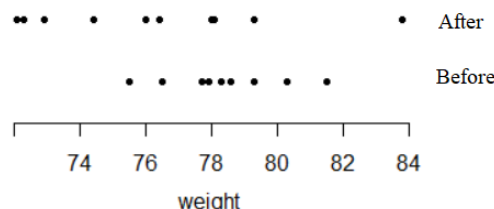
The first male student in the sample paid €92.18 and the first female student in the sample paid €108.52. But unlike our weight loss program example there is no relationship or pairing between these two students.

Visualising the data

A clustered dot plot is shown for the data opposite.

It does show a tendency for the after weights to be lower than the before weights.

But such a plot does not show which before weight corresponds to which after weight.



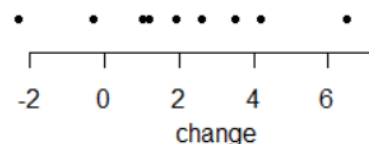
To better view the data calculate the weight change (i.e. before – after) and plot this instead.

Before	77.9	76.5	78.3	81.5	79.3	80.3	78.6	77.7	75.5	77.9
After	74.4	72.3	76.4	83.8	78.1	79.3	72.1	78	72.9	76
Change	3.5	4.2	1.9	-2.3	1.2	1	6.5	-0.3	2.6	1.9

A dot plot of the difference (or change) is shown opposite.

We can see that most of the data (all but two values) are positive – indicating that most subjects reduced their weight, suggesting that the weight loss program is effective.

(The clustered dot plot above does not show this as effectively)



The paired t test

The paired sampled t tests should be used instead of independent sample t tests when there is a pairing between the first and second sample.

The null hypothesis will be that the difference from before to after is zero, the alternative is that it is not zero.

I.e. letting μ_{before} = true mean weight before the program and similarly define μ_{after} then

$H_0: \mu_{\text{before}} = \mu_{\text{after}}$

$H_0: \mu_{\text{before}} \neq \mu_{\text{after}}$

We can invoke a paired t test in R by specifying the additional argument, `paired = TRUE`

```
> t.test(before, after, paired=TRUE)

Paired t-test

data:  before and after
t = 2.6289, df = 9, p-value = 0.02741
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 0.2818152 3.7581848
sample estimates:
mean of the differences
                2.02
```

The output indicates that the mean weight loss was 2.02 Kg and a 95% confidence interval estimate for weight loss is given by [0.28, 3.76] kg.

As this excludes zero, we can (subject to a normality check) conclude that there is a statistically significant difference between the before and after weights.

Normality?

The paired t test assumes that the difference between the two samples is normally distributed.

```
shapiro.test(change)

Shapiro-Wilk normality test

data:  change
W = 0.98562, p-value = 0.9881
```

As the p value > 0.05 we conclude that the data is normal and (if the data was suitably collected), we can conclude that there is a statistically significant weight loss.

Could the (independent samples) t test be applied?

We could have addressed this question by ignoring the pairing structure and applied an unpaired, or independent samples t test

```
> t.test(before, after)

Welch Two Sample t-test

data:  before and after
t = 1.5787, df = 12.855, p-value = 0.1387
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.7474114  4.7874114
sample estimates:
mean of x mean of y
   78.35    76.33
```

Note that with this approach, we failed to establish a statistically significant result. If there is paired structure in the data, then usually the paired test is more powerful than the independent samples test

(More powerful means that it is more likely that a statistically significant result will be found)

Investigators can exploit this by being clever in the design of their experiments.

Example

Two electrical screw drivers J and K are being compared to determine how quickly each can sink an 8 cm screw in to timber blocks, with each screw driver to be tested 10 times.

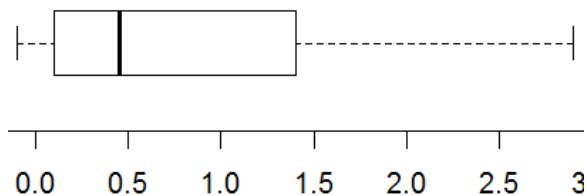
A less efficient approach here might be to take 20 timber blocks, using half as test units for J and the other half for K. The only analysis that could be done in this case would be the independent samples t -test.

Instead, a total of just 10 block samples are selected for the investigation and two screws are inserted into each piece, one with each screwdriver. The data (seconds) is reproduced below.

Sample	1	2	3	4	5	6	7	8	9	10
Brand J	13.8	19.6	24.3	19.5	19.2	23.3	22.7	20.2	16.0	25.1
Brand K	13.4	19.1	22.9	19.4	17.3	23.4	19.8	20.1	15.6	24.6

A boxplot of the difference between brand J – brand K data gives a clear indication (note that the plot is not centred around zero) that brand J is taking longer, and therefore, not as good as brand K.

```
boxplot (brandJ-brandK, horizontal=TRUE)
```



This is confirmed by the output of the paired samples t test (we omit the stating of the null and alternative hypothesis – but if we elect to always use a two-sided test, then there is no ambiguity)

```
t.test (brandJ,brandK, paired=TRUE)
```

Paired t-test

```
data: brandJ and brandK
t = 2.6723, df = 9, p-value = 0.02553
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 0.1243097 1.4956903
sample estimates:
mean of the differences
      0.81
```

As with the previous example, an independent samples test would have failed to establish a difference.

We frequently can engineer a way to modify an experiment to induce a paired structure in it, as was done in the last problem.

For example, to see if two techniques for measuring the calorie content of a food product give different measures, we might split the experimental units (food items) in two and randomly assign one half each to the two methods

If it were not convenient, or possible, to split the experimental units and if a number of technicians were carrying out the experiments, we might direct each technician to carry out two measures – one with each technique.

Or, if experiments were time-consuming and say, only two experiments could be done per day, we might use the day of the experiment to induce a pairing.

Or, in medical investigations, patients in a treatment and control group may be matched. I.e. if there are two subjects of the same sex, same smoking status, similar age etc. then one may be assigned to the treatment group and the other to the control group. Ideally, the choice as to which goes to which group should be decided by random assignment.

Exercise

Below are some of the variables used in the customer survey data.

Variable	Records
q1_sex	Male or female
q2_age	Age
q3_goods	Type of goods purchased (electronics, jewellery, fragrances)
q4_spend	Amount spent by customer
q5_delivery	Time delay from order to delivery
q6_paymode	One of credit card, debit card, PayPal or other
q7_isCustomer	Is the customer an existing customer (yes, no)
q8_lastSpend	If yes to Q7, how much was their last spend?
q9_lastDelivery	If yes to Q7, how long did their last delivery take?

Identify two examples where each of the following statistical tests might be used and write the necessary R line of code to run each test.

- Independent samples t test
- Paired samples t test

In each case, carry out the test and check the normality assumptions and thus, determine if a statistically significant finding can be established.